YUVAINTERN · EMBEDDED SYSTEMS ENGINEER INTERNSHIP

SmartSense

Battery-Powered Smart Home Environment Monitor

Week 1 Task: System Requirements Analysis and Block Diagram Design

Item	Details
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Abstract

This report defines the requirements and architecture of SmartSense, a battery-powered smart home sensor node that measures temperature, relative humidity, barometric pressure, ambient light and room occupancy, and reports the data over Wi-Fi to a home MQTT broker. It contains a structured specification of 54 numbered requirements, a hardware block diagram, a layered software architecture, an operating-cycle flowchart, a component selection study with a weighted decision matrix, a power budget estimating 69 days of operation per charge at a five-minute reporting interval, a requirements traceability matrix, a risk register and a verification plan for the remaining internship weeks.

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1	Introduction	Background, problem statement, objectives, scope, stakeholders, link to the 4-week plan
2	Methodology	Approach, requirement conventions, assumptions, information sources, work breakdown
3	System Requirements Analysis	Use cases, functional, performance, I/O, interface, power, real-time and non-functional requirements
4	System Design	Hardware block diagram, component selection, interfaces, power budget, software architecture, data model, traceability, risks, verification plan
5	Conclusions	Summary, compliance with objectives, limitations and next steps
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A	Appendix A	Indicative bill of materials
B	Appendix B	Glossary of abbreviations

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Figure 2	Layered software architecture
Figure 3	Operating-cycle flowchart
Figure 4	Estimated current during one active phase
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1. Introduction

1.1 Background and motivation

Smart-home sensing is one of the most common applications of low-power embedded systems. A sensor node must run for months on a small battery, take trustworthy measurements, communicate reliably over a wireless link and remain inexpensive. In such products the hardware choices (microcontroller, sensors, battery, radio) fix most of the firmware design space and cannot be changed cheaply once boards are built. Defining requirements and architecture carefully before writing any code is therefore the most valuable step in the development cycle, and it is the focus of this first internship task.

1.2 Problem statement

Occupants of a home usually have little visibility of indoor conditions. Persistently high humidity encourages mould growth, temperature differences between rooms cause discomfort and wasted heating or cooling energy, lights are left on in unoccupied rooms, and many commercial sensors depend on a proprietary cloud service. The problem addressed here is to specify and design an **open, battery-powered, Wi-Fi-connected multi-sensor node** that reports room temperature, humidity, pressure, light level and motion to a home automation system using the open MQTT protocol, while lasting at least two months on a single battery charge.

1.3 Project objectives

1. Define a clear and testable set of system requirements covering functionality, performance, inputs and outputs, communication interfaces, power management and real-time behaviour.
2. Design a complete hardware block diagram and a software architecture that show every major component and how they interact.
3. Select components through documented trade-offs and justify each choice against the requirements.
4. Demonstrate through calculation (power budget, I2C bus timing, regulator headroom) that the design is feasible before any hardware is built.
5. Provide traceability and a verification plan so that Weeks 2 to 4 (firmware, debugging, integration and testing) can start directly from this document.

1.4 Scope

In scope	Out of scope
Requirements specification and analysis	PCB layout and schematic capture
Hardware and software architecture	Enclosure mechanical design
Component selection and trade-offs	Cloud or smartphone application development
Power, timing and bus-level feasibility analysis	Production regulatory certification
Risk register and verification plan	Home-automation rules (these run on the broker side)

SmartSense is a hypothetical device. No hardware is built in Week 1; all figures are derived from manufacturer datasheets and engineering analysis, and are flagged as estimates that the later weeks will validate by simulation and test.

1.5 Stakeholders

Stakeholder	Interest	Key needs
Home occupant	Comfort, energy saving, healthy air	Accurate readings, long battery life, easy set-up, no cloud lock-in
Installer / maintainer	Fast deployment and support	Simple provisioning, over-the-air updates, clear status LED, diagnostics
Firmware developer	Maintainable code	Modular architecture, documented interfaces, testable requirements
Home-automation platform	Data consumer	Standard MQTT topics, JSON payload, predictable timing

1.6 Relationship to the four-week internship plan

Week	Task	How this document supports it
1	Requirements and block diagram	This report: requirements, architecture, design rationale
2	Firmware development	Software architecture, pin map, timing constraints and data model define what to code (timer interrupts, UART console, sensor reading, LED control)
3	Debugging and optimisation	Power budget and real-time constraints give measurable targets for optimisation
4	Integration and final test report	Traceability matrix and verification plan (V-01 to V-10) become the test suite

2. Methodology

2.1 Overall approach

The work followed a top-down requirements-engineering process aligned with ISO/IEC/IEEE 29148 [11], followed by architecture partitioning and feasibility analysis:

1. **Problem definition.** Identify the application, the stakeholders and the operating environment.
2. **Use-case analysis.** Describe the normal and exceptional scenarios the device must handle (Section 3.2).
3. **Requirement derivation.** Convert scenarios and stakeholder needs into numbered, testable statements grouped by category (Section 3).
4. **Architecture partitioning.** Split the system into hardware blocks and software layers and define the interfaces between them (Section 4).
5. **Component selection.** Compare candidate parts using a weighted decision matrix and datasheet limits, then record the rationale (Section 4.4).
6. **Feasibility analysis.** Check the design with calculations: energy budget, I2C rise time, regulator headroom and memory layout (Sections 4.5 to 4.9).
7. **Traceability and verification planning.** Map every requirement to a design element and a verification method (Sections 4.10 to 4.12).

2.2 Requirement conventions

Convention	Meaning
shall / should / could	Mandatory / strongly desired / optional. The Priority column uses Must (M), Should (S) and Could (C) following the MoSCoW method.
Requirement IDs	FR = functional, PR = performance, PW = power, RT = real-time, NF = non-functional. Each ID is unique and is referenced in the traceability matrix.
Verification method	T = test (measurement), A = analysis (calculation), I = inspection (design review), D = demonstration.
Units	SI units; currents in mA or µA; energy per cycle expressed as charge in mA·s; battery life in days.

2.3 Assumptions and constraints

Ref	Assumption or constraint
A1	Indoor residential use with an existing 2.4 GHz Wi-Fi network and an MQTT broker (for example Mosquitto or Home Assistant) on the local network.
A2	Power source is a single protected Li-ion cell (3.7 V nominal, 2000 mAh) that can be recharged through USB-C. Usable capacity is taken as 85 % (1700 mAh) between 4.2 V and 3.3 V.
A3	Self-discharge and battery ageing are not included in battery-life estimates; they are treated as a margin item for the verification phase.
A4	Current figures for Wi-Fi phases are engineering estimates based on typical manufacturer data; they are validated in Weeks 3 and 4.
A5	Firmware is written in C on ESP-IDF (FreeRTOS). In the internship, hardware is simulated where a physical board is not available.
C1	Bill of materials target of US$20 or less at prototype quantities.
C2	Enclosure no larger than 70 × 50 × 25 mm.

2.4 Information sources

Design decisions were based on manufacturer datasheets for the ESP32-C3 module [1][2], SHT40 [4], BMP280 [5], BH1750 [6], MCP73831 [7] and AP2112 [8]; the I2C-bus specification [9]; the MQTT 3.1.1 standard [10]; and ESP-IDF documentation [3]. Block diagrams and charts were produced with Python and Matplotlib so that all drawings and calculated values come from one consistent set of numbers.

2.5 Work breakdown and effort

Activity	Output	Hours
Domain research and benchmarking of existing sensor nodes	Application definition, candidate technologies	4
Stakeholder needs and use-case analysis	Use-case table (Section 3.2)	3
Requirements specification	54 numbered requirements (Section 3)	6
Component selection and trade-off study	Decision matrix and rationale tables (Section 4.4)	5
Hardware block diagram design	Figure 1 and signal-flow description	3
Software architecture and operating logic	Figures 2 and 3, module table, data model	3
Power, timing and bus feasibility analysis	Budget tables, Figures 4 and 5	3
Traceability, risks, verification plan, review and report writing	Sections 4.10 to 5, appendices	5
Total		32

3. System Requirements Analysis

3.1 Product overview and intended functionality

SmartSense is a small wall- or shelf-mounted node placed in a room. It spends almost all of its time in deep sleep. Every five minutes (configurable) it wakes, measures temperature, relative humidity, barometric pressure, ambient light and battery voltage, joins the home Wi-Fi network, publishes a JSON message to an MQTT broker over TLS and returns to sleep. A passive infrared (PIR) sensor wakes the node immediately when motion is detected so that occupancy changes are reported within seconds. First-time set-up is done from a smartphone over Bluetooth LE, and firmware can be updated over the air.

Table 3.1: Key performance targets at a glance

Parameter	Target
Measured quantities	Temperature, relative humidity, pressure, illuminance, motion, battery voltage
Default reporting interval	300 s (configurable 60 s to 3600 s)
Measurement accuracy (system level)	±0.5 °C, ±3 %RH, ±2 hPa, ±20 % illuminance
Wake-to-publish time	≤ 5 s (design value 3.0 s)
Deep-sleep current (whole board)	≤ 100 µA
Battery life, 300 s interval	≥ 60 days (estimated 69.1 days)
Data delivery	≥ 99 % of scheduled reports at RSSI ≥ −70 dBm

Table 3.1 (continued): Overview of the 54 requirements by category

Category	ID range	Count	Must	Should	Could
Functional	FR-01 to FR-19	19	13	5	1
Performance	PR-01 to PR-10	10	4	6	0
Power	PW-01 to PW-08	8	7	1	0
Real-time (soft / hard, Table 3.9)	RT-01 to RT-08	8	n/a	n/a	n/a
Non-functional	NF-01 to NF-09	9	6	3	0
Total		54	30	15	1

3.2 Operating scenarios (use cases)

ID	Scenario	Trigger	Expected system behaviour
UC-1	Routine monitoring	RTC timer (300 s)	Wake, measure all sensors and battery, connect, publish telemetry, sleep.
UC-2	Motion detected	PIR rising edge	Wake, read sensors, publish a motion event within 5 s (at most one per 30 s), sleep.
UC-3	First-time set-up	No configuration stored, or button held 5 s	Start BLE provisioning (LED fast blink, 5-minute time-out), receive Wi-Fi and broker settings, store them in NVS, restart.
UC-4	Network outage	Wi-Fi or broker unreachable	Abort within time-outs, store the reading in an RTC-memory buffer (up to 48), send buffered readings after the next successful connection.
UC-5	Low battery	VBAT ≤ 3.5 V, then ≤ 3.3 V	Set low-battery flag and LED pattern; at 3.3 V publish a final alert and shut down until the button is pressed and VBAT is at least 3.5 V.
UC-6	Firmware update	OTA command on the command topic, VBAT ≥ 3.7 V	Download image over HTTPS, verify, boot new slot, confirm or roll back.
UC-7	Charging	USB-C connected	Charger recharges the cell at up to 500 mA; the node continues its normal cycle.

3.3 Functional requirements

Table 3.2: Functional requirements (Pri = priority, Ver = verification method)

ID	Requirement	Pri	Ver
FR-01	The system shall measure ambient temperature with the SHT40 sensor on every measurement cycle.	M	T
FR-02	The system shall measure relative humidity with the SHT40 sensor on every measurement cycle.	M	T
FR-03	The system shall measure barometric pressure with the BMP280 sensor on every measurement cycle.	M	T
FR-04	The system shall measure ambient illuminance with the BH1750 sensor on every measurement cycle.	M	T
FR-05	The system shall detect motion with a PIR sensor and raise a motion event, including while in deep sleep.	M	T, D
FR-06	The system shall measure battery voltage and report an estimated state of charge in percent.	M	T
FR-07	The system shall wake periodically from deep sleep and report all measurements at the interval defined in PR-07.	M	T
FR-08	The system shall publish telemetry as a JSON message using MQTT 3.1.1 over TLS (port 8883) with QoS 1.	M	T
FR-09	The system shall publish a motion event within the latency of RT-03, limited to one event per 30 s.	M	T
FR-10	The system shall support BLE provisioning of Wi-Fi credentials, broker address and credentials, and room name, started by a 5 s button press or when no configuration exists.	M	D
FR-11	The system shall store its configuration in non-volatile storage (NVS) so that it survives resets and power loss.	M	T
FR-12	The system shall buffer at least 48 readings in RTC memory while the network is unavailable and send them in order after reconnection.	S	T

ID	Requirement	Pri	Ver
FR-13	The system shall support over-the-air firmware update over HTTPS with automatic rollback if the new image fails validation.	S	D
FR-14	The system shall indicate boot, provisioning, connecting, error and low-battery states with distinct status-LED patterns.	S	I, D
FR-15	Each message shall include diagnostic fields: firmware version, reset reason, RSSI and status flags.	S	T
FR-16	The system shall detect sensor faults (no acknowledge, CRC error, out-of-range value), flag them in the message and continue with the remaining sensors.	M	T
FR-17	The system shall synchronise its clock by SNTP and time-stamp messages in UTC.	S	T
FR-18	The system shall warn at VBAT ≤ 3.5 V and shut down at VBAT ≤ 3.3 V to protect the cell and keep the supply rail in range.	M	T
FR-19	The system should provide a USB serial console for logs and diagnostics.	C	D

3.4 Performance requirements

Table 3.3: Performance requirements

ID	Requirement	Pri	Ver
PR-01	Temperature accuracy shall be within ±0.5 °C over 10 °C to 40 °C (the SHT40 itself is typically ±0.2 °C).	M	T
PR-02	Relative humidity accuracy shall be within ±3 %RH over 20 %RH to 80 %RH at 25 °C (SHT40 typically ±1.8 %RH).	M	T
PR-03	Pressure accuracy shall be within ±2 hPa over 950 hPa to 1050 hPa (BMP280 typically ±1 hPa absolute).	S	T
PR-04	Illuminance shall be reported from 1 lx to 65535 lx with ±20 % accuracy.	S	T
PR-05	Reported resolution shall be 0.1 °C, 0.1 %RH, 0.1 hPa, 1 lx and 0.01 V (battery).	S	I
PR-06	At least 99 % of scheduled reports shall be delivered (PUBACK received) over 24 h at RSSI ≥ −70 dBm.	M	T
PR-07	The reporting interval shall be configurable from 60 s to 3600 s with a default of 300 s.	M	T
PR-08	The node shall operate reliably down to RSSI = −75 dBm, using retries within the time-outs of RT-04.	S	T
PR-09	Message time stamps shall be within ±2 s of UTC after SNTP synchronisation.	S	T
PR-10	A telemetry message shall not exceed 256 bytes.	S	I

3.5 Input and output specification

Table 3.4: Inputs

Input	Device	Range	Accuracy / resolution	Interface	Trigger
Temperature	SHT40	−40 to 125 °C (device)	±0.2 °C typ., 0.01 °C	I2C 0x44	Each cycle
Relative humidity	SHT40	0 to 100 %RH	±1.8 %RH typ., 0.01 %RH	I2C 0x44	Each cycle
Pressure	BMP280	300 to 1100 hPa	±1 hPa typ.	I2C 0x76	Each cycle (forced mode)
Illuminance	BH1750	1 to 65535 lx	±20 %	I2C 0x23	Each cycle (one-time high-resolution mode)
Motion	AM312 PIR	Several metres	Digital (binary)	GPIO3	Edge interrupt, wakes from deep sleep
Battery voltage	Divider ÷ 2 + ADC1	3.0 to 4.2 V (ADC sees 1.5 to 2.1 V)	±50 mV	GPIO1	Each cycle, 8-sample average
User button	Tactile switch	n/a	30 ms debounce; 5 s long press	GPIO4	Edge interrupt, wakes from deep sleep
Settings and commands	BLE / MQTT command topic	n/a	n/a	Radio	On demand

Table 3.5: Outputs

Output	Destination / interface	Content	When
Telemetry message	MQTT topic …/telemetry, QoS 1	JSON, about 200 bytes: temperature, humidity, pressure, lux, motion, battery, RSSI, flags	Every reporting interval
Motion event	MQTT topic …/event/motion, QoS 1	JSON with time stamp and sensor snapshot	On PIR event (max 1 per 30 s)
Status message	MQTT topic …/status, retained	Firmware version, reset reason, uptime counters	After every boot
Status LED	GPIO10 (through 330 Ω)	Patterns defined in Table 4.11	State changes
Debug log	USB serial console	Timestamped log lines	Continuous when connected

3.6 Communication interfaces

Table 3.6: Communication interfaces

Interface	Specification	Used for
I2C	Fast-mode 400 kHz [9], 7-bit addresses 0x23, 0x44, 0x76, 4.7 kΩ pull-ups to 3.3 V, SDA = GPIO6, SCL = GPIO7	SHT40, BMP280, BH1750
Wi-Fi	IEEE 802.11 b/g/n, 2.4 GHz, WPA2-PSK or WPA3-SAE, DHCP, fast reconnect using saved channel and BSSID	Telemetry, OTA, SNTP
Bluetooth LE	Bluetooth 5 LE, GATT provisioning service, encrypted session protected by a proof-of-possession code printed on the label	First-time set-up only
MQTT	OASIS MQTT 3.1.1 [10] over TLS 1.2, port 8883, QoS 1, persistent session so commands are queued while the node sleeps	Telemetry, events, commands
HTTPS	TLS 1.2 with server certificate validation	Firmware download (OTA)
SNTP	RFC 4330 [13]	Time synchronisation
USB	USB 2.0 full speed through the ESP32-C3 USB Serial/JTAG controller; VBUS also feeds the charger	Flashing, console, charging
GPIO / ADC	3.3 V logic; 12-bit SAR ADC (ADC1)	PIR, button, LED, battery sense

Table 3.7: Example telemetry payload (about 200 bytes)

```
{"id":"ss-
a1b2c3","fw":"1.0.0","ts":1792400000,"temp_c":24.6,"rh_pct":51.3,"press_hpa":1009.2,"lux
":142,"motion":false,"vbat_v":3.87,"soc_pct":62,"rssi_dbm":-58,"flags":0}
```

3.7 Power management requirements

Table 3.8: Power requirements

ID	Requirement	Pri	Ver
PW-01	The node shall be powered by one 3.7 V Li-ion cell (2000 mAh) that is charged through USB-C; it shall also be flashable and debuggable over the same connector.	M	I
PW-02	Estimated battery life at the default 300 s interval shall be at least 60 days.	M	A, T
PW-03	Total board current in deep sleep shall not exceed 100 µA.	M	T
PW-04	The charge consumed by one active phase shall not exceed 300 mA·s.	M	T
PW-05	A full charge shall complete within 6 hours.	S	T
PW-06	The 3.3 V rail shall stay within 3.0 V to 3.6 V (module operating range) during 350 mA radio bursts for all battery voltages of at least 3.3 V.	M	A, T
PW-07	The cell shall be protected against over-charge, over-discharge and short circuit (protection circuit module, PCM).	M	I
PW-08	Battery voltage shall be measured to within ±50 mV.	M	T

3.8 Real-time and timing constraints

SmartSense is a soft real-time system: missing a deadline degrades service or battery life but is not hazardous. Time-outs are nevertheless enforced in firmware so that a fault can never keep the radio on and drain the battery.

Table 3.9: Real-time requirements

ID	Constraint	Value	Type
RT-01	Sensor acquisition from trigger to all three values available	≤ 400 ms (the BH1750 conversion of up to 180 ms dominates; sensors are triggered together)	Soft
RT-02	Wake-up to publish acknowledged	≤ 5 s (design value 3.0 s)	Soft
RT-03	PIR edge to motion event acknowledged	≤ 5 s	Soft
RT-04	Network time-outs	Wi-Fi connect 10 s, MQTT publish 5 s, then abort the cycle and sleep	Hard (firmware enforced)
RT-05	Watchdogs	Task watchdog 10 s; whole active phase capped at 20 s	Hard
RT-06	Button handling	30 ms debounce, 5 s long press, provisioning time-out 5 min	Soft
RT-07	Reporting-period accuracy	±5 % of the configured interval (internal RTC oscillator drift)	Soft
RT-08	Interrupt service routines	Shorter than 50 µs; they only post an event to a FreeRTOS queue	Hard

3.9 Non-functional requirements

Table 3.10: Non-functional requirements

ID	Requirement	Pri	Ver
NF-01	All network traffic that carries data or credentials shall use TLS 1.2 or higher with server certificate validation.	M	T, I
NF-02	No credentials shall be hard-coded; they shall be supplied only through provisioning and stored in NVS.	M	I
NF-03	The design shall support secure boot and flash encryption for production units.	S	I
NF-04	Any firmware lock-up shall recover automatically through the watchdog within 20 s.	M	T
NF-05	The node shall operate indoors from 0 °C to 45 °C and 10 % to 90 % RH, non-condensing.	M	T
NF-06	The enclosure shall not exceed 70 × 50 × 25 mm and shall provide a PIR window and ventilation openings for the humidity sensor.	S	I
NF-07	Bill of materials cost shall not exceed US$20 at prototype quantities.	S	I
NF-08	A pre-certified radio module shall be used, and the Li-ion cell shall comply with IEC 62133-2 [12].	M	I
NF-09	Firmware shall be modular C with documented interfaces and shall not exceed 1.2 MB, so that two OTA images fit in flash.	M	I, A

4. System Design

4.1 Architecture overview and design drivers

Five design drivers shaped the architecture: (1) very low sleep current, because battery life depends mainly on the 297 s spent asleep in every 300 s cycle; (2) integrated Wi-Fi and Bluetooth LE, to avoid a separate radio and a hub; (3) measurement accuracy, which favours digital sensors with factory calibration; (4) open protocols (MQTT, I2C, USB), to avoid vendor lock-in; and (5) low cost and a small board, which favours a single pre-certified module.

The resulting architecture is a **single-MCU star topology**. The ESP32-C3 module is the only bus master; three digital sensors are I2C slaves on one shared bus; the PIR, button and LED use dedicated GPIOs; and a small power subsystem converts USB or battery power into a clean 3.3 V rail and reports battery voltage through the ADC. The firmware is event-driven: the node stays in deep sleep and is woken by the RTC timer, the PIR or the button.

Figure 1 on the next page shows the complete hardware architecture. The subsequent subsections explain each block, each connection and the reasons for the choices.

4.2 Hardware block diagram

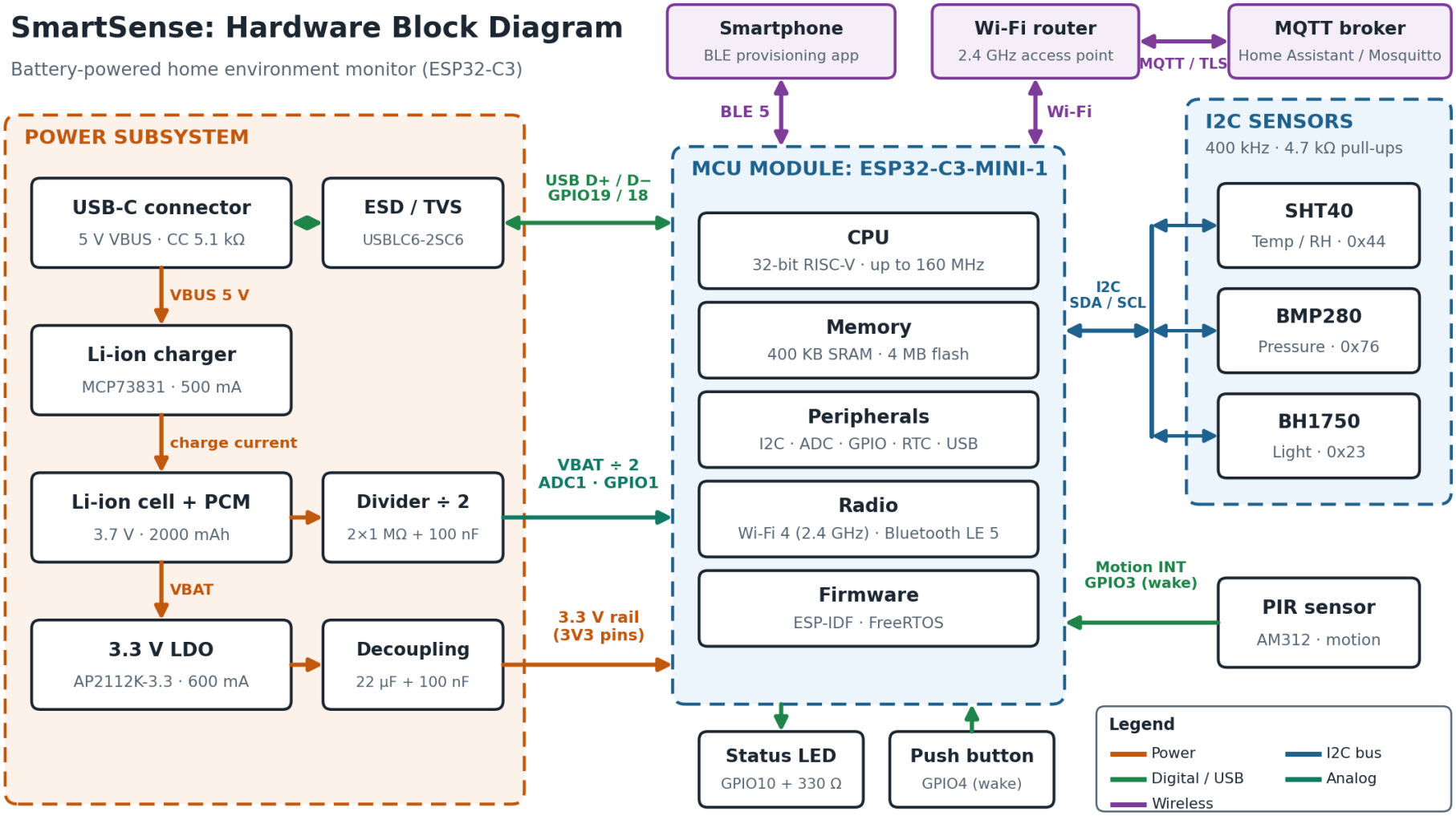


Figure 1: SmartSense hardware block diagram showing power, I2C, digital, analog and wireless connections.

4.3 Signal and power flow

The diagram can be read as four independent flows:

1. **Power flow.** USB-C delivers 5 V VBUS to the MCP73831 charger, which charges the protected Li-ion cell at up to 500 mA. The cell (3.0 V to 4.2 V) feeds the AP2112K-3.3 low-dropout regulator, which supplies the 3.3 V rail to the module, sensors and PIR. Local decoupling (22 µF and 100 nF) absorbs the radio current bursts.
2. **Measurement flow.** The three I2C sensors share SDA and SCL. The MCU triggers all conversions together and then reads the results, so total acquisition time is set by the slowest sensor (BH1750). The PIR output is a simple digital level that drives an RTC-capable GPIO and can wake the chip from deep sleep.
3. **Battery monitoring.** A 2 × 1 MΩ divider scales the cell voltage (maximum 4.2 V) to 2.1 V for ADC1 on GPIO1. A 100 nF capacitor lowers the source impedance seen by the ADC, and the divider draws only about 2 µA.
4. **Wireless flow.** Telemetry leaves the module over Wi-Fi to the home router and then to the MQTT broker, which feeds the dashboard and automations. Bluetooth LE is used only for provisioning from a smartphone. USB carries firmware, logs and charging power.

A typical cycle therefore reads: timer wake, power up the sensors, trigger conversions, read and validate the data, measure the battery, connect to Wi-Fi, publish over MQTT, confirm delivery and return to deep sleep (Figure 3).

4.4 Component selection and justification

4.4.1 Microcontroller module

Four candidate platforms were compared with a weighted decision matrix. Scores run from 1 (poor) to 5 (excellent) and reflect the needs of this project based on datasheet information; they are engineering judgements, not benchmark results.

Table 4.1: Weighted decision matrix for the microcontroller

Criterion (weight)	ESP32-C3-MINI-1	ESP32-S3-MINI-1	nRF52840 module	STM32L4 + Wi-Fi module
Integrated Wi-Fi and BLE (25)	5	5	2 (BLE only)	3
Deep-sleep power (20)	4	4	5	5
Toolchain and ecosystem (15)	5	5	4	4
Cost (15)	5	4	3	2
Peripherals: I2C, ADC, USB (10)	4	5	5	5
Certified module and size (15)	5	5	4	3
Weighted score (out of 100)	94	93	73	72

The **ESP32-C3-MINI-1** [2] is selected. It integrates a 32-bit RISC-V core running at up to 160 MHz, 400 KB of SRAM, 4 MB of flash, 2.4 GHz Wi-Fi, Bluetooth 5 LE, a 12-bit ADC and a USB Serial/JTAG controller in a 13.2 × 16.6 mm certified module, with a deep-sleep current of about 5 µA at chip level [1]. The ESP32-S3 scores almost the same but adds a second core and resources that this application does not need at higher cost. The nRF52840 has no Wi-Fi, which would require a hub. The STM32L4 option needs an external Wi-Fi module, which increases cost and board area.

4.4.2 Sensors

Table 4.2: Sensor selection

Function	Selected	Alternatives considered	Reason for selection
Temperature and humidity	SHT40 (Sensirion)	DHT22; BME280	Typical accuracy ±0.2 °C and ±1.8 %RH, I2C with CRC-8, about 8 ms conversion and sub-µA standby. The DHT22 uses a slow proprietary one-wire protocol; the BME280 is cheaper per function but its self-heating and ±3 %RH humidity accuracy are weaker.
Pressure	BMP280 (Bosch)	LPS22HB; BME280	±1 hPa absolute accuracy, 0.1 µA in sleep, forced mode for one-shot readings, supply 1.71 V to 3.6 V.
Ambient light	BH1750 (ROHM)	VEML7700; photoresistor with ADC	Digital lux output with a spectral response close to the human eye, 1 lx to 65535 lx, power-down current about 0.01 µA, no ADC channel needed.
Motion	AM312 mini PIR	mmWave radar; ultrasonic	Draws about 15 µA continuously and gives a clean digital output for wake-up; radar modules draw tens of mA, which would destroy battery life.

I2C addresses are unique (0x44, 0x76, 0x23), so no address conflicts exist on the shared bus. All three sensors run from 3.3 V.

4.4.3 Power components

Table 4.3: Power component selection

Block	Selected part	Key data	Reason
Battery	1S Li-ion, 2000 mAh, with PCM	3.7 V nominal, 4.2 V full, about 3.0 V empty	High energy density, rechargeable, protected against abuse (PW-07).
Charger	MCP73831	4.2 V CC/CV, programmable up to 500 mA, STAT output; 2 kΩ on PROG sets 500 mA	Small SOT-23-5 part that needs few external components. Full charge takes about 4 to 5 hours.
Regulator	AP2112K-3.3	600 mA, dropout about 250 mV at full load, quiescent current about 55 µA	Low-noise LDO for RF, simple and cheap. A buck-boost converter would be more efficient at low battery voltage but costs more and needs a larger board.
USB protection	USBLC6-2SC6 + 5.1 kΩ CC resistors	ESD protection on D+/D−; resistors identify the node as a USB sink	Protects the module against ESD and lets any USB-C source supply 5 V.
Battery sense	2 × 1 MΩ + 100 nF	Scales 4.2 V to 2.1 V, about 2 µA	Meets ±50 mV accuracy with ADC calibration; low leakage.

4.4.4 Communication technology

Table 4.4: Communication choices

Decision	Chosen	Alternatives	Rationale
Radio	Wi-Fi (plus BLE for set-up)	Zigbee / Thread, BLE-only, LoRa	Works with the existing router and needs no hub or gateway. The cost is higher radio energy, which is managed by duty cycling (Section 4.6).
Application protocol	MQTT 3.1.1 over TLS	HTTP REST, CoAP	Lightweight publish/subscribe with QoS levels and persistent sessions; supported natively by Home Assistant and most brokers.
Sensor bus	I2C	SPI, 1-Wire	Two wires for three devices, standard speed, low power; all chosen sensors support it.
Data format	JSON	CBOR, binary	Human-readable, easy to debug and consumed directly by dashboards; 200 bytes is acceptable.

4.5 Interface and pin design

Table 4.5: Proposed pin assignment (ESP32-C3-MINI-1)

Signal	GPIO	Direction	Notes
I2C SDA	GPIO6	Bidirectional	4.7 kΩ pull-up to 3.3 V
I2C SCL	GPIO7	Output (open drain)	4.7 kΩ pull-up to 3.3 V
Battery sense	GPIO1 (ADC1_CH1)	Analog input	12 dB attenuation, eFuse calibration
PIR output	GPIO3	Input	RTC-capable, wake source from deep sleep
Push button	GPIO4	Input, active low	RTC-capable wake source; 10 kΩ external pull-up
Status LED	GPIO10	Output	330 Ω series resistor, about 4 mA
USB D− / D+	GPIO18 / GPIO19	USB	Native USB Serial/JTAG, protected by USBLC6-2SC6
Reset (EN)	EN pin	Input	RC reset network and optional reset button

The pins were chosen to avoid the ESP32-C3 strapping pins (GPIO2, GPIO8 and GPIO9), which determine boot mode. Only GPIO0 to GPIO5 can wake the chip from deep sleep, so the PIR (GPIO3) and button (GPIO4) are placed there.

I2C electrical check

For Fast-mode (400 kHz) the I2C specification [9] limits the rise time to 300 ns. With pull-up resistance Rp = 4.7 kΩ and an estimated bus capacitance Cb = 60 pF (three devices plus traces), the rise time is tr = 0.8473 × Rp × Cb = 0.8473 × 4700 Ω × 60 pF ≈ **239 ns**, which meets the limit with margin. The minimum allowed pull-up for 3 mA sink current is (3.3 V − 0.4 V) / 3 mA ≈ 970 Ω, so 4.7 kΩ also keeps the low-level current small (about 0.7 mA only while a line is held low).

4.6 Power architecture and energy budget

4.6.1 Deep-sleep current budget

Table 4.6: Deep-sleep current budget

Contributor	Current (µA)	Source / note
ESP32-C3 chip in deep sleep (RTC timer and RTC memory on)	5	ESP32-C3 datasheet [1]
AP2112K-3.3 quiescent current	55	AP2112 datasheet [8]
PIR AM312 (always on)	15	Manufacturer typical value
Battery divider (4.2 V across 2 MΩ)	2	Calculated: 4.2 V / 2 MΩ ≈ 2.1 µA
SHT40, BMP280, BH1750 in sleep or power-down	1	Datasheets [4][5][6], combined
Margin for charger leakage, pull-ups and board leakage	22	Engineering allowance
Total sleep budget	100	Meets PW-03 (≤ 100 µA)

4.6.2 Active-phase energy

Table 4.7: Active-phase charge (estimated)

Phase	Duration (s)	Average current (mA)	Charge (mA·s)
Boot + init	0.3	30	9
Sensor read	0.3	30	9
Wi-Fi connect	1.0	110	110
TLS + MQTT publish	1.2	120	144
Shutdown	0.2	30	6
Total active phase	3.0	92.7 (average)	278

The Wi-Fi radio can draw peaks of roughly 350 mA during transmission [1]; Table 4.7 lists time-averaged phase currents, which determine battery life. The peaks matter for regulator and decoupling design and are covered in Section 4.6.4.

4.6.3 Battery life

Average current over one reporting period T is **I_avg = (Q_active + I_sleep × (T − t_active)) / T** and battery life is **L = C_usable / I_avg**, with C_usable = 1700 mAh (85 % of 2000 mAh), Q_active = 278 mA·s, t_active = 3.0 s and I_sleep = 0.1 mA. For the default T = 300 s: I_avg = (278 + 0.1 × 297) / 300 = 1.026 mA and L = 1700 / 1.026 ≈ 1657 h ≈ **69.1 days**, which exceeds PW-02 (60 days) by about 15 %.

Table 4.8: Estimated battery life for different reporting intervals

Reporting interval	Average current (mA)	Estimated battery life (days)	Comment
1 min	4.728	15.0	Too short for the 60-day goal
5 min	1.026	69.1	Default; meets PW-02
15 min	0.409	173.4	Meets PW-02
30 min	0.254	278.6	Self-discharge becomes the limit (A3)
60 min	0.177	399.9	Self-discharge becomes the limit (A3)

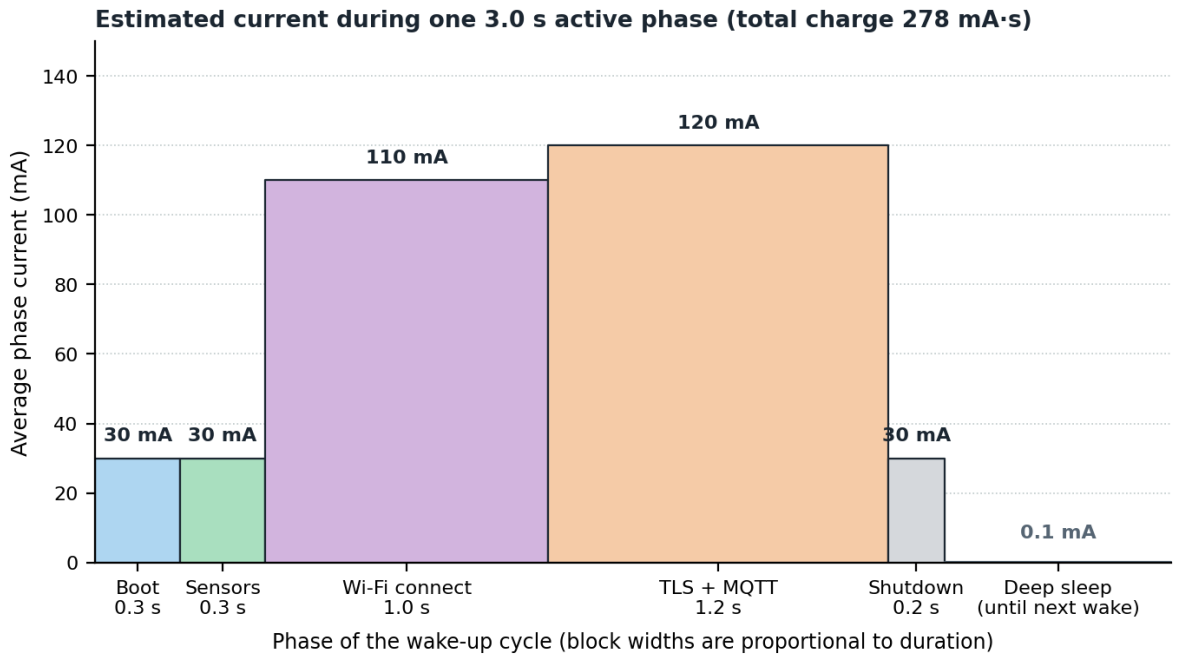


Figure 4: Estimated current during one 3.0 s active phase.

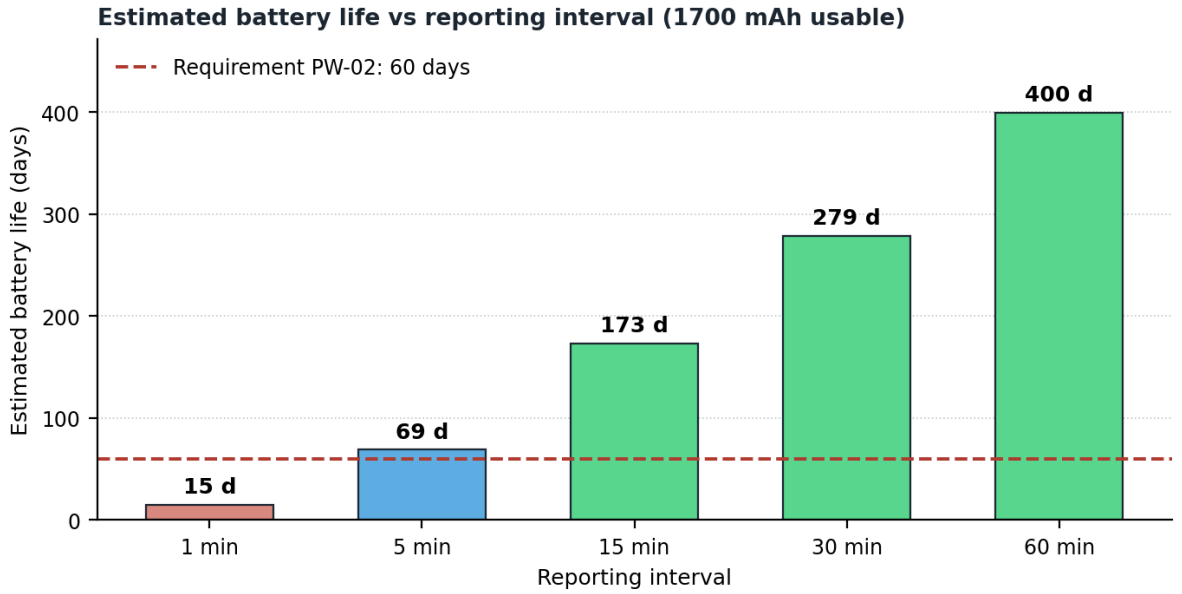


Figure 5: Estimated battery life versus reporting interval, compared with the 60-day requirement.

4.6.4 Supply headroom during radio bursts

At the shutdown threshold VBAT = 3.3 V, a 350 mA burst causes a cell voltage drop of roughly 50 mV (assuming 100 to 150 mΩ internal resistance) and an LDO dropout of about 250 mV × (350 / 600) ≈ 150 mV. The rail therefore stays near 3.1 V, above the 3.0 V minimum of the module (PW-06). This is why the firmware shuts down at 3.3 V instead of using the full cell capacity, and why 22 µF bulk plus 100 nF decoupling is placed next to the module. The linear regulator wastes (VBAT − 3.3 V) / VBAT of the energy, about 11 % at 3.7 V; this is accepted for simplicity, low RF noise and cost.

4.7 Software architecture and operating logic

The firmware follows a three-layer structure (Figure 2). The **application layer** contains six FreeRTOS-based modules with a single responsibility each; the **middleware layer** consists of standard ESP-IDF components [3]; and the **driver layer** isolates the hardware so that sensors or pins can change

without touching application logic. The operating cycle is described by the flowchart in Figure 3. Both figures follow on the next two pages.

Table 4.9: Application modules

Module	Responsibility	Related requirements
Main controller	Handles wake cause, runs the state machine (BOOT, MEASURE, TRANSMIT, SLEEP, PROVISION, FAULT), feeds the watchdogs and enforces the 20 s cycle cap.	FR-07, RT-04, RT-05, NF-04
Sensor manager	Initialises sensors, triggers conversions together, reads results, checks CRC and range, retries three times, sets fault flags, receives PIR events from the ISR queue.	FR-01 to FR-05, FR-16, RT-01, RT-08
Power manager	Reads battery voltage (8-sample average), converts to percent with a lookup table, applies 3.5 V and 3.3 V thresholds, configures deep-sleep wake sources, manages the RTC-memory buffer.	FR-06, FR-12, FR-18, PW-03, PW-08
Connectivity manager	Fast Wi-Fi reconnect, SNTP, MQTT connect/publish/subscribe with time-outs and back-off, flushes buffered readings.	FR-08, FR-09, FR-12, FR-17, PR-06
Provisioning and UI	BLE provisioning service, button debounce and long-press detection, status LED patterns.	FR-10, FR-14, RT-06
OTA and config manager	NVS read/write of settings, HTTPS OTA with dual slots, marks new image valid or rolls back.	FR-11, FR-13, NF-02, NF-03

4.7.1 Software architecture diagram

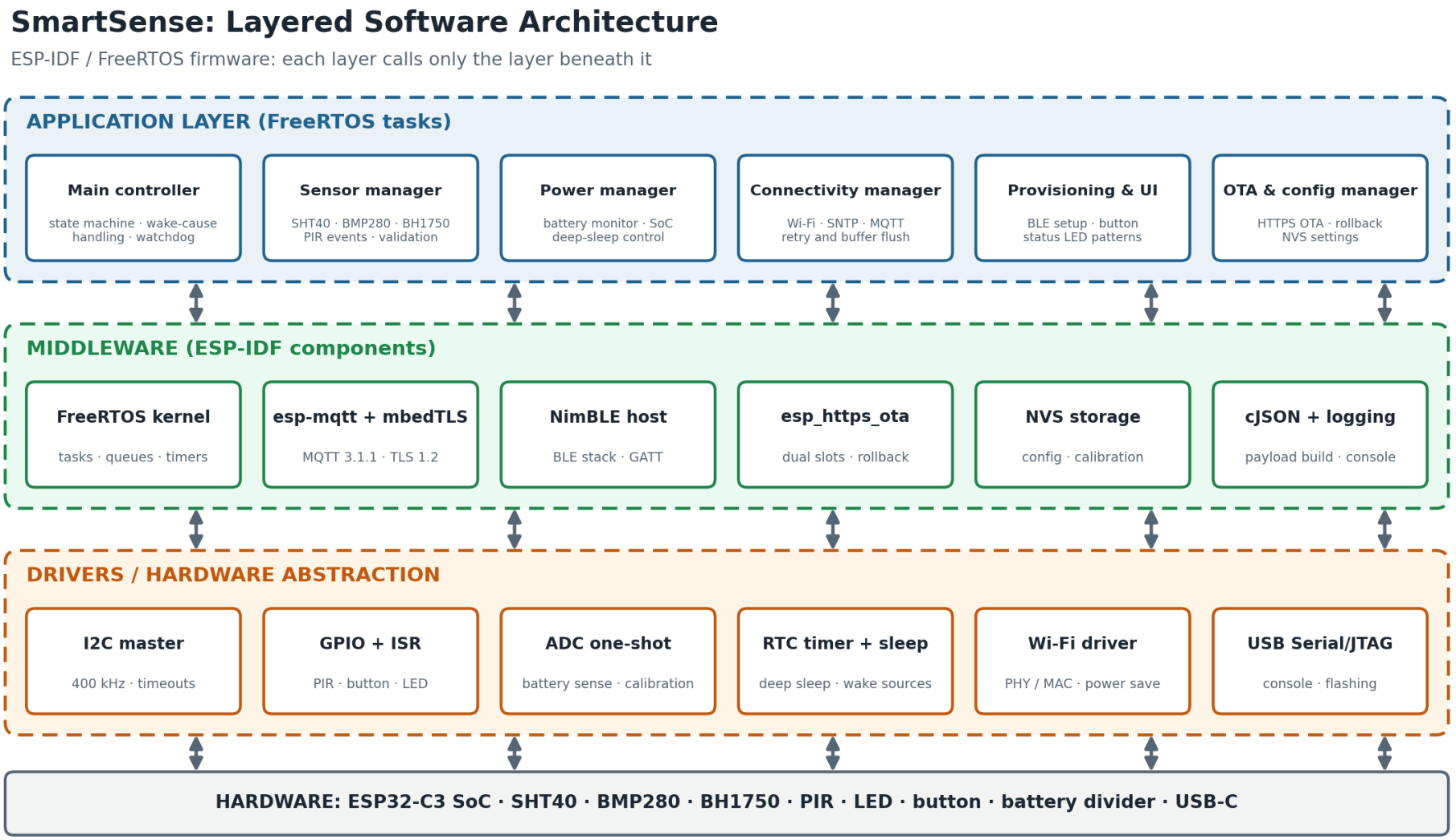


Figure 2: Layered software architecture of the SmartSense firmware.

4.7.2 Operating-cycle flowchart

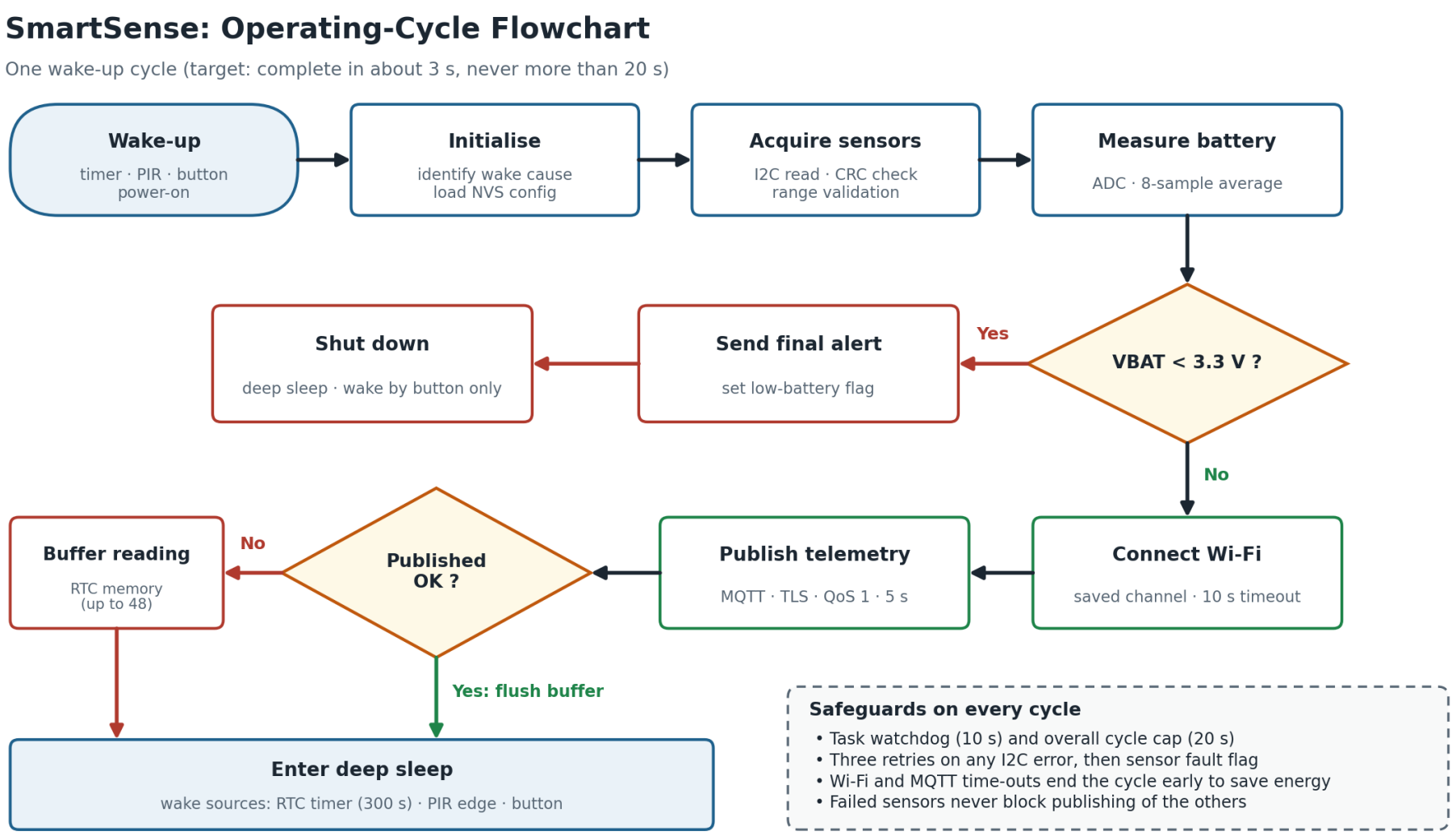


Figure 3: Operating-cycle flowchart for one wake-up, including low-battery and network-failure paths.

4.8 Operating modes

Table 4.10: Operating modes

Mode	Description	Typical current	Entry	Exit
Active	Measure and transmit, about 3 s per cycle	About 93 mA average (peaks about 350 mA)	Timer, PIR or button wake	Cycle complete or time-out
Deep sleep	CPU off; RTC timer, RTC memory and wake GPIOs on	≤ 0.1 mA (whole board)	End of cycle	RTC timer, PIR edge, button
Provisioning	BLE advertising, LED fast blink, 5-minute limit	Tens of mA	No config or 5 s button press	Config saved or time-out
OTA update	Wi-Fi connected, image download and verification	About 100 to 120 mA	OTA command and VBAT ≥ 3.7 V	Reboot into new image
Critical-battery shutdown	Deep sleep with timer disabled	≤ 0.1 mA	VBAT ≤ 3.3 V	Button press and VBAT ≥ 3.5 V

Table 4.11: Status LED patterns (single LED on GPIO10)

State	Pattern
Boot	One 100 ms flash
Provisioning	Fast blink, 5 Hz
Connecting to Wi-Fi or broker	One short flash per second
Error (sensor or network fault)	Three flashes, then off
Low battery	Two short flashes at the end of each cycle

4.9 Data model and storage design

4.9.1 MQTT topics

Table 4.12: MQTT topic structure (home/<room>/<device_id>/...)

Topic suffix	Direction	QoS	Content
telemetry	Publish	1	Periodic JSON measurement message (Table 3.7)
event/motion	Publish	1	Motion event with time stamp
status	Publish, retained	1	Firmware version, reset reason, uptime counters, battery state
cmd	Subscribe	1	Interval change, OTA trigger, reboot. Queued by the broker through a persistent session and delivered while the node is awake.

4.9.2 Status flag field

Bit	Meaning
0	SHT40 fault
1	BMP280 fault
2	BH1750 fault
3	Low battery (VBAT ≤ 3.5 V)
4	Buffered (late) reading

4.9.3 Flash partition plan (4 MB)

Table 4.13: Flash partition table

Name	Offset	Size	Purpose
nvs	0x009000	24 KB	Configuration, credentials, calibration
otadata	0x00F000	8 KB	Active OTA slot and rollback state
phy_init	0x011000	4 KB	Radio calibration data
ota_0	0x020000	1.5 MB	Application image slot A
ota_1	0x1A0000	1.5 MB	Application image slot B
coredump	0x320000	64 KB	Crash dump for post-mortem debugging
storage	0x330000	832 KB	Event log and spare data

The partitions end exactly at 0x400000 (4 MB). The firmware target of 1.2 MB (NF-09) fits in a 1.5 MB slot with 20 % margin. The RTC memory buffer holds 48 readings of 16 bytes each (768 bytes) out of the 8 KB of RTC memory, which covers 4 hours of outage at the default interval.

4.10 Requirements traceability matrix

Table 4.14: Traceability from requirements to design elements and verification

Requirements	Design element	Verification
FR-01 to FR-04, PR-01 to PR-05	SHT40, BMP280, BH1750 on I2C; Sensor manager	V-01
FR-05, FR-09, RT-03, RT-08	AM312 PIR, GPIO3 wake, ISR-to-queue design	V-08
FR-06, FR-18, PW-06, PW-08	Battery divider, ADC1, Power manager thresholds, LDO headroom	V-04
FR-07, PR-07, RT-07	RTC timer wake, Main controller	V-02, V-03
FR-08, PR-06, PR-08, PR-10, NF-01	Wi-Fi, MQTT over TLS, Connectivity manager	V-03
FR-10, FR-11, NF-02	BLE provisioning, NVS, Provisioning and UI	V-06
FR-12	RTC-memory ring buffer (48 × 16 B)	V-07
FR-13, NF-03	Dual OTA slots with rollback, secure boot option	V-05
FR-14, FR-15, FR-16, FR-17, FR-19	LED patterns, diagnostics fields, fault flags, SNTP, USB console	V-09
PW-01, PW-05, PW-07	USB-C, MCP73831, protected cell	V-10
PW-02, PW-03, PW-04	Deep-sleep design, energy budget (Tables 4.6 to 4.8)	V-02
RT-01, RT-02, RT-04, RT-05, NF-04	Time-outs, task watchdog, 20 s cycle cap	V-02, V-03
NF-05 to NF-09	Enclosure, certified module, BOM and firmware size review	Inspection and analysis

4.11 Risk analysis

Table 4.15: Risk register (L = likelihood, I = impact; H/M/L)

ID	Risk	L	I	Mitigation
R1	Radio current bursts cause brown-out resets	M	H	22 µF + 100 nF decoupling at the module, low-ESR LDO, shutdown at 3.3 V (Section 4.6.4).
R2	Self-heating and board heat bias the temperature reading	M	M	Place the SHT40 away from the module and charger, add thermal slots, sample once and power down, apply an offset determined in test.
R3	Wi-Fi access point unavailable	M	M	Time-outs, back-off and RTC-memory buffer for 48 readings (FR-12).

ID	Risk	L	I	Mitigation
R4	I2C bus lock-up	L	H	Three retries, bus-clear procedure (up to 9 SCL pulses), fault flag, task watchdog.
R5	PIR false triggers shorten battery life	M	M	30 s rate limit, ignore events during the connect phase, sensitivity tuning.
R6	Failed OTA update leaves the node unusable	L	H	Dual slots, rollback if the new image is not confirmed, battery ≥ 3.7 V condition.
R7	Li-ion safety, including charging in the cold	L	H	Protected certified cell (IEC 62133-2), 500 mA charge limit, indoor use; production variant should use a charger with NTC temperature input.
R8	RTC oscillator drift	H	L	Accepted (±5 %, RT-07); time stamps come from SNTP on each cycle.
R9	Credential leakage	L	H	TLS with certificate validation, encrypted provisioning session, option for NVS and flash encryption (NF-03).
R10	Part shortage	M	M	Footprint-compatible alternatives: SHT41 or BME280, other 3.3 V LDOs.

4.12 Verification plan

The verification tests below will be executed in simulation or on a prototype during Weeks 2 to 4, and form the basis of the final test report.

Table 4.16: Verification tests

ID	Test	Method	Pass criterion
V-01	Sensor accuracy	Compare with a calibrated reference thermo-hygrometer; use saturated salt solutions ($MgCl_2$ about 33 %RH, NaCl about 75 %RH) for humidity points	PR-01 to PR-04
V-02	Current profile and battery life	Measure with a current profiler or shunt and oscilloscope over several cycles; compute average current	Sleep ≤ 100 µA; Q ≤ 300 mA·s; life ≥ 60 days; wake-to-publish ≤ 5 s
V-03	24-hour network soak	Run at 300 s interval and count acknowledged messages; record RSSI and watchdog resets	At least 286 of 288 messages; no unrecovered lock-up
V-04	Low-battery and brown-out	Ramp supply from 4.2 V to 3.3 V under radio bursts with a bench supply	Rail ≥ 3.0 V; warning at 3.5 V; shutdown at 3.3 V
V-05	OTA and rollback	Update with a good image, then with a deliberately faulty image	Good image runs; faulty image rolls back
V-06	Provisioning	Provision a blank node from a phone; power-cycle; try a wrong password	Settings persist; errors reported by LED and console
V-07	Outage resilience	Switch off the router for 60 minutes, then restore	All 12 readings delivered in order with the late flag set
V-08	Motion latency	Trigger the PIR 20 times at varied intervals	Latency ≤ 5 s; at most one event per 30 s
V-09	Interface and diagnostics demonstration	Observe LED patterns, console log, diagnostic fields and fault flags	Behaviour matches Tables 4.11 and 3.5
V-10	Charging	Charge from 3.3 V to full at 500 mA	Complete within 6 h; termination at 4.20 V ± 1 %

5. Conclusions

This report defined the requirements and designed the architecture of SmartSense, a battery-powered smart home environment monitor. The result is a specification of 54 numbered, testable requirements; a hardware block diagram and a layered software architecture; documented trade-offs for the microcontroller, sensors, power components and communication technologies; and feasibility calculations covering energy, I2C timing and supply headroom.

Table 5.1: How the objectives were met

Objective	How it was met	Where
Clear, testable requirements	54 numbered requirements with priority and verification method	Section 3
Complete block diagrams	Hardware block diagram, software layers and operating-cycle flowchart	Figures 1 to 3
Justified component choices	Weighted decision matrix and rationale tables for sensors, power and communication	Section 4.4
Proven feasibility	Sleep budget, active energy, battery life of 69.1 days (goal 60), I2C rise time 239 ns (limit 300 ns), rail headroom	Sections 4.5 and 4.6
Traceability and verification	Traceability matrix, risk register and ten verification tests	Sections 4.10 to 4.12

5.1 Limitations

- The device is hypothetical. Active-phase currents, bus capacitance and enclosure effects are estimates and must be confirmed by measurement.
- Battery-life estimates exclude self-discharge, battery ageing and cold-temperature capacity loss.
- Weighted scores in the decision matrix are engineering judgements based on datasheets.

5.2 Next steps

- **Week 2:** implement firmware modules in C (sensor sampling, timer-interrupt scheduler, status LED, UART console) following Sections 4.7 and 4.9, and simulate hardware where needed.
- **Week 3:** debug and optimise against the budgets in Section 4.6 and the timing limits in Table 3.9.
- **Week 4:** integrate the modules and run tests V-01 to V-10 as the final test report.

References

[1] Espressif Systems, ESP32-C3 Series Datasheet.

[2] Espressif Systems, ESP32-C3-MINI-1 and ESP32-C3-MINI-1U Datasheet.

[3] Espressif Systems, ESP-IDF Programming Guide (sleep modes, OTA updates, provisioning, NVS, partition tables).

[4] Sensirion AG, SHT4x Datasheet: Digital Humidity and Temperature Sensor (SHT40).

[5] Bosch Sensortec, BMP280 Digital Pressure Sensor Datasheet.

[6] ROHM Semiconductor, BH1750FVI Digital 16-bit Serial Output Type Ambient Light Sensor IC Datasheet.

[7] Microchip Technology, MCP73831/2 Miniature Single-Cell, Fully Integrated Li-Ion/Li-Polymer Charge Management Controllers Datasheet.

[8] Diodes Incorporated, AP2112 600 mA CMOS LDO Regulator Datasheet.

[9] NXP Semiconductors, UM10204: I2C-bus specification and user manual.

[10] OASIS Standard, MQTT Version 3.1.1, 2014.

[11] ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes: Requirements engineering.

[12] IEC 62133-2, Safety requirements for portable sealed secondary lithium cells and batteries for use in portable applications.

[13] IETF RFC 4330, Simple Network Time Protocol (SNTP) Version 4.

Appendix A: Indicative bill of materials

Ref	Part	Function	Key specification
U1	ESP32-C3-MINI-1 (4 MB flash)	MCU and radio	RISC-V 160 MHz, Wi-Fi 4, BLE 5, certified module
U2	SHT40	Temperature and humidity	±0.2 °C, ±1.8 %RH typ., I2C 0x44
U3	BMP280	Pressure	±1 hPa typ., I2C 0x76
U4	BH1750FVI	Ambient light	1 to 65535 lx, I2C 0x23
U5	AM312	PIR motion sensor	About 15 µA, digital output
U6	MCP73831	Li-ion charger	4.2 V, 500 mA with 2 kΩ PROG
U7	AP2112K-3.3	3.3 V LDO	600 mA, Iq about 55 µA
U8	USBLC6-2SC6	USB ESD protection	Low-capacitance TVS array
BT1	1S Li-ion 2000 mAh with PCM	Battery	3.7 V nominal, IEC 62133-2 compliant
J1	USB-C receptacle (with 2 × 5.1 kΩ on CC)	Power and programming	USB 2.0
R, C	Pull-ups, divider, LED resistor, decoupling	Passive components	4.7 kΩ ×2, 1 MΩ ×2, 330 Ω, 2 kΩ, 22 µF, 100 nF
D1, SW1	Status LED, tactile switch	User interface	GPIO10, GPIO4

Appendix B: Glossary

Term	Meaning
ADC	Analog-to-digital converter
BLE	Bluetooth Low Energy
BOM	Bill of materials
CC/CV	Constant current / constant voltage (charging profile)
GPIO	General-purpose input/output
I2C	Inter-integrated circuit serial bus
LDO	Low-dropout linear regulator
MQTT	Message Queuing Telemetry Transport
NVS	Non-volatile storage
OTA	Over-the-air (firmware update)
PCM	Protection circuit module (battery)
PIR	Passive infrared (motion sensor)
RSSI	Received signal strength indicator
RTC	Real-time clock and low-power domain
SNTP	Simple Network Time Protocol
TLS	Transport Layer Security